

Jacob Yen

San Francisco, CA | jyen6432@gmail.com | (628) 225-8368 | [linkedin.com/in/jacob-yen](https://www.linkedin.com/in/jacob-yen) | github.com/KooptaTroopta

SUMMARY

Motivated computer engineering student with a solid academic and computer science foundation. Rapid learner and dependable team member/leader. Passionate about delivering high quality projects and constantly growing and improving.

EDUCATION

Boston University College of Engineering, Boston, MA

May 2028

Bachelor of Science in Computer Engineering | GPA: 4.00

Honors & Activities: Dean's List, HackHardware, Society of Asian Scientists and Engineers, Intramural Basketball

Relevant Coursework: Software Engineering, Applied Algorithms (In Progress), Prob. & Statistics, Engineering Design

PROJECTS

Social Network Graph App | C++, GitHub

Nov - Dec 2025

- Led a team of 5 computer engineering students on a network visualization app, collaborated using Agile project management
- Collaborated using GitHub Actions, via approved PRs, Kanban pipeline, applied CI/CD workflows, and automated unit tests
- Designed and implemented a person similarity model, analyzing data from profiles, to simulate a social network of 63 students
- Integrated Qt-based GUI with C++ backend for profile visualization, graph traversal, and pathfinding (breadth-first search)
- Added automatic dataset saving, intuitive dataset editing, and random dataset generation for user customization

Interactive Truss Simulation and Optimization Software | MATLAB

Nov - Dec 2025

- Created MATLAB program to compute static internal forces, failure loads, and graphed designs for trusses
- Optimized existing design using Monte Carlo simulation, achieving 72.6% increased load above minimum capacity, 8.88% additional capacity from preliminary model design, 7.7% improved load-cost ratio, and strongest valid truss design in class
- Implemented buckling-strength regression, geometry and constraint verification, and GUI to automate improvement process
- Constructed acrylic truss using optimal model, compared calculated and physical results by measuring uncertainty

Automatic Soap Dispenser | Arduino

Sep - Dec 2025

- Managed a team of 4 engineers to design, prototype, and manufacture a compact dispenser, applying industry design practices
- Designed and implemented an Arduino-based contactless liquid dispenser for a client, integrating infrared sensing, algorithms for pump control, and LCD/OLED UI for animated water level display and current device status feedback
- Configured embedded firmware to ensure reliable activation (95% detection accuracy) and consistent 25 mL dispensing

Custom-Built Temperature Monitoring Device | Arduino, Onshape

Apr - May 2025

- Engineered temperature monitoring device through the full product design lifecycle to ensure dependable operation
- Developed embedded software displaying real-time temperature readings on an LCD screen and implemented user-configurable temperature thresholds with visual and audible alerts when readings fall outside the specified range
- Programmed on Arduino, modeled components in Onshape, and fabricated parts using 3D printing and hand soldering

PROFESSIONAL EXPERIENCE

Freelance Web Development

Aug - Oct 2025

- Built and launched an event website with online payments and registration links, driving a 30% increase in sign-ups
- Improved SEO and site visibility, boosting traffic by ~40% and expanded reach via QR codes and shareable marketing links
- Drove 25% higher event attendance, increased recurring revenue, improved customer experience and community engagement

LEADERSHIP & ACTIVITIES

Peer Tutor (Informal), Boston, MA

Sep 2024 - Present

- Provide academic support to 10+ Boston University students through frequent study sessions and homework help
- Guide students through complex technical concepts for engineering mechanics, physics, statistics, and software engineering
- Assess peer skill level and customize study plans, resulting in intuitive understanding and higher course performance

SKILLS

- **Programming Languages:** C++, C, MATLAB, Python (pandas, NumPy, Matplotlib), Java, JavaScript, HTML, CSS
- **Technologies:** Git/GitHub, Arduino, Google Suite, Microsoft Office, Onshape CAD, Squarespace
- **Languages & Interests:** Chinese (Proficient) | Pickup Basketball, League of Legends, Teaching, 3D Modeling/Rendering